REMEDIATION PLAN

VIRTUAL CONTROL

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# NSX Transport Node VDS Reconciliation

Recovering the four host transport nodes after a vCenter redeploy changed the VDS UUID.

NSX | TRANSPORT NODES | VDS UUID | VCENTER REDEPLOY | REMEDIATION

v1.0 VMware Cloud Foundation 9

# NSX Transport Node VDS Reconciliation Plan

**Prepared by:** Virtual Control LLC **Date:** June 9, 2026 **Document Version:** 1.0 **Classification:** Internal

**Scope.** All four host transport nodes (esxi01–esxi04) are in `failed` state because NSX references a **VDS UUID that no longer exists** after the vCenter redeploy / VDS re-import. This plan reconciles the NSX transport-node host-switch to the **current** VDS so the data plane realizes.

**Read this first:** NSX is **already in a degraded, interlocked state** in this environment (also missing from SDDC Manager inventory; vCenter shows `ACTIVATING` in SDDC). This is **not a quick fix** — execute only in a **maintenance window**, **after an NSX Manager backup**, and **not during the Aria workshop**. Several execution steps must be **confirmed against the live NSX 9.0.x UI/API before running** — do not run them blind.

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## 1. Confirmed Diagnosis

Verified live on 2026-06-09 via the NSX API and vCenter:

**All four host transport nodes fail identically**, with NSX retrying indefinitely (220+ retries):

> `failure_code 8804 / 9538` — **"NSX Ownership Validation failed. Cannot find the HostSwitch [50 37 13 3f 96 e4 3e 93-1d f0 3b 01 d4 76 6a f8] of type VDS."**

| Transport Node | Host | State |
|---|---|---|
| 7a93666c-... | esxi01.lab.local | failed |
| fa720366-... | esxi02.lab.local | failed |
| 87d46d7c-... | esxi03.lab.local | failed |
| e491fa06-... | esxi04.lab.local | failed |

**The UUID mismatch (the smoking gun):**

| | VDS UUID |
|---|---|
| **NSX transport nodes expect** | `50 37 13 3f 96 e4 3e 93-1d f0 3b 01 d4 76 6a f8` |
| **Actual VDS in vCenter** ( `vcenter-cl01-vds01` , 4 hosts, v9.0.0) | `50 12 35 19 fa f5 be 0a-71 8d ab e0 b5 e6 b9 cf` |

Per-node sub-states confirm it: `HostConfig` and `LogicalSwitchFullSync` = **success**, but `Host` (ownership) and `Start` = **failed** on the missing VDS. NSX management and control clusters are **STABLE**; this is purely a host-switch/VDS reference problem.

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## 2. Why It Happened

- The vCenter was **redeployed** and the VDS **re-imported** (5/13 work). A re-created/re-imported VDS gets a **new UUID**.
- NSX prepared these hosts originally against the **old** VDS UUID and **stored that reference** in the transport-node / transport-node-profile (TNP) host-switch spec.
- NSX validates that it "owns" that exact VDS UUID before realizing the data plane. The old UUID is gone, so validation fails and loops forever.
- Same root family as **NSX missing from SDDC Manager inventory** and **vCenter `ACTIVATING`** — all downstream of the vCenter redeploy.

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## 3. Impact

- **NSX data plane (overlay) is not realized** on any host — overlay segments / TEP networking won't function.
- **Aria workshop Day 2** (NSX on-demand networks, security groups, LB) **will fail** until fixed. *Workaround for the workshop:* use a Network Profile with an **existing VLAN-backed** network and defer the NSX on-demand exercises.
- Management/control plane and VLAN-backed networking are unaffected.

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## 4. Hard Constraints

1. **NSX Manager backup is mandatory before any change** — this is the rollback path. Configure/verify a backup (SFTP) and take an on-demand backup first.
2. **Maintenance window only**, not during the workshop. Reconciling transport nodes can briefly disrupt host networking.
3. **NSX is already tangled** (missing from SDDC inventory). Fixing the VDS reference may not fully restore SDDC-Manager↔NSX inventory — that is a separate item.
4. **No support contract** — so this is self-service; proceed conservatively, one cluster, verify each step.
5. **Confirm exact UI/API steps against the live NSX 9.0.x build before executing.** The mechanism below is correct; specific labels/payloads must be verified, not assumed.
6. **Fragile single-host nested lab** — snapshot/backup, change one thing at a time, verify NSX + vSAN health between steps.

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## 5. Pre-Flight (Read-Only)

```
 # 1) Re-confirm the per-node failure + the expected (stale) VDS UUID
curl -sk -u 'admin:<nsx-pw>' \
  https://nsx-vip.lab.local/api/v1/transport-nodes/<TN-ID>/state | python -m json.tool

# 2) Confirm the CURRENT VDS UUID in vCenter (the target to reconcile to)
#    PowerCLI: Get-VDSwitch | Select Name, @{n='UUID';e={$_.ExtensionData.Uuid}}
#    Expected: vcenter-cl01-vds01 = 50 12 35 19 fa f5 be 0a-71 8d ab e0 b5 e6 b9 cf
```

- [ ] **NSX compute manager (vCenter) registration is healthy** and NSX can see the **current** VDS (System → Fabric → Compute Managers → vCenter = *Registered / Up*; re-sync if needed). NSX can only map the TNP to a VDS it discovers from the compute manager. *(History: NSX↔vCenter CM registration has needed repair after redeploy — error 90348 / certificate-management service. Resolve CM registration first if not green.)*
- [ ] NSX **management + control clusters STABLE** (already confirmed).
- [ ] **Take an NSX Manager backup** (System → Backup & Restore → Backup now). Record the timestamp.
- [ ] Record current TNP name and the cluster (compute collection) it's applied to.
- [ ] vSAN healthy, resync 0 (don't reconcile during storage work).

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## 6. Remediation

### Approach A — Reconcile the Transport Node Profile to the current VDS (primary)

Goal: point NSX's host-switch at `vcenter-cl01-vds01` (current UUID) and re-apply, so the four TNs re-realize.

1. **System** → **Fabric** → **Hosts** → **Clusters** — locate `vcenter-cl01` and the applied **Transport Node Profile**.
2. Edit (or create a new) **Transport Node Profile** whose **host switch maps to the current VDS** (`vcenter-cl01-vds01`). Keep the same transport zones, uplink profile, and TEP IP pool as the existing config.

3. **Re-apply / update** the TNP to the cluster. NSX re-validates ownership against the **new** VDS UUID and re-realizes each host.
4. Watch each transport node move `failed → in_progress → success`.

> **Confirm before running:** the exact 9.0.x path for editing the host-switch VDS mapping and whether your hosts were prepared via a **cluster-level TNP** vs **individual transport nodes**. If individual, the equivalent is updating each transport node's host-switch spec ( `PUT /api/v1/transport-nodes/<id>` with the corrected `host_switch_id` ) — verify the payload against `GET` of the current object first; never PUT a hand-built body.

### Approach B — Detach and re-prepare NSX on the cluster (fallback)

Use only if Approach A cannot bind to the new VDS (e.g., stale ownership won't clear).

1. **Remove NSX (detach transport nodes)** from the cluster — clears the stale host-switch ownership.
2. Confirm hosts are clean (no residual NSX VIBs/host-switch errors).
3. **Re-prepare** the cluster with a TNP mapped to the **current** VDS.
4. Re-realize and verify.

> Approach B is more disruptive (full host re-prep) — prefer A. Related references: KB 319037, KB 374279, KB 317923.

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## 7. Verification

- [ ] All four transport nodes report **state = success** ( `GET /api/v1/transport-nodes/<id>/state` ).
- [ ] **Transport nodes: 4/4 healthy** (NSX UI System → Fabric → Hosts; and re-run the VCF Health Check).
- [ ] No `8804/9538` ownership errors; no retry loop.
- [ ] **Data-plane test:** create a test overlay segment, attach a test VM, confirm TEP/overlay connectivity (then remove).
- [ ] vSAN + cluster still healthy.
- [ ] Re-check whether NSX now re-appears in **SDDC Manager inventory** (may still require separate remediation).

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## 8. Rollback

1. If transport nodes do not recover or networking regresses, **restore the NSX Manager backup** taken in Pre-Flight (System → Backup & Restore → Restore).
2. If Approach B was used and re-prep failed, the hosts remain on the VLAN-backed VDS for management; restore NSX backup and re-engage the plan after root-causing the bind failure.

> Rollback is the NSX Manager backup — which is why Pre-Flight makes it mandatory.

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## 9. Risk & Caveats

| Risk | Mitigation |
| --- | --- |
| Brief host networking disruption during re-apply | Maintenance window; one cluster; mgmt on VLAN-backed portgroups unaffected |
| Stale ownership won't clear (A fails) | Fallback to Approach B (detach + re-prepare) |
| CM registration not actually healthy → NSX can't see current VDS | Pre-Flight gate: fix compute-manager registration first |
| Fix doesn't restore SDDC-Manager inventory | Tracked separately; this plan targets the data plane only |
| Unverified API payloads break NSX further | Never PUT hand-built bodies; GET-modify-PUT; verify 9.0.x steps live first |

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## 10. Decision

> **Decision:** In a maintenance window, after an NSX Manager backup, reconcile the host Transport Node Profile to the **current** VDS ( `vcenter-cl01-vds01` , UUID `50 12 35 19…` ) and re-apply to `vcenter-cl01` (Approach A); fall back to detach-and-re-prepare (Approach B) only if ownership won't rebind. **Justification:** The transport nodes fail solely because they reference a VDS UUID that the vCenter redeploy invalidated; re-pointing them at the live VDS is the minimal, prescribed-mechanism fix. **Implication:** Requires a window + NSX backup; exact 9.0.x steps confirmed live before execution; does not by itself guarantee SDDC-Manager inventory recovery.

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## 11. Document Information

| Field | Value |
| --- | --- |
| Title | NSX Transport Node VDS Reconciliation Plan |
| Document ID | VC-PLAN-NSX-TN-VDS-RECON-20260609 |
| Version | 1.0 |
| Issued | 2026-06-09 |
| Author | Virtual Control LLC |
| Classification | Internal |
| Platform | VMware Cloud Foundation 9.0.1 (nested) — NSX on VDS |
| Applies to | Host transport nodes esxi01–esxi04 failed with VDS-UUID ownership mismatch |
| Related Documents | VCF9-SDDC-Manager-Inventory-State-Known-Issues-2026-06-08; VCF-VDS-Reimport-Runbook-2026-05-13; VCF9-Host-vCenter-Rebuild-Plan |
| Distribution | Lab notebook |

## Revision History

| Version | Date | Notes |
| --- | --- | --- |
| 1.0 | 2026-06-09 | Initial plan. Confirmed diagnosis (TN VDS-UUID mismatch after vCenter redeploy: NSX expects `50 37 13 3f…`, current VDS is `50 12 35 19…`); impact; hard constraints (NSX backup, window, confirm-live); read-only pre-flight; remediation Approach A (reconcile TNP to current VDS + re-apply) and Approach B (detach + re-prepare); verification; rollback via NSX backup; risk register; decision. |

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**Document End**